Who Audits the Audit?

What we did, what we found, and why it matters

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1 THE QUESTION

Fairness audits are becoming gates — written into procurement, bias-audit regulation, and model sign-off, so that systems ship or do not ship on what an audit reports. But an audit is software, and software has defaults. We asked what those defaults decide on our behalf.

2 WHAT WE DID

A complete fairness audit over three lending decisions: consumer credit (German Credit), mortgage origination (HMDA Georgia), peer-to-peer default risk (Lending Club). A deterministic AIF360 layer computes every metric and assigns every severity label by fixed rule; a language model supplies only interpretation and is scored against that fixed ground truth, so no model grades another model. Audits run over the **full cleaned population** (1,000 and 10,978 rows), every row scored out-of-fold by a model that never saw it. Then we audited the audit. Frozen prompts, offline replay, a $15 cap; **$0.89 spent over 98 calls**, 60 tests green.

3 WHAT WE FOUND

Table 1: Each component is defensible in isolation. Each silently resolves a normative question.

Component	What it decided
Reporting floor ($n \geq 15$)	Suppressed **5 of 11** HMDA race×sex cells. Audit reported MOD. (0.8169); worst real disparity **0.1917**. All five suppressed were racial minorities.
Age threshold	40+ is employment law (ADEA); ECOA protects 62+. Banded age reads 0.9791 (LOW); statutory cut reads **0.8897** — the largest age disparity in the data.
Favorable-label convention	Computing on the adverse label of `loan_default` gave DI = **0.0** (CRITICAL) where the true value is **0.9931**.
Refusal detector (<40 chars)	6 flagged, **0** genuine. Zeroed 14 cycles of a second provider, moving its mean 70.0 → 19.79 — and that provider is **permanently blocked** in our config on that evidence.
Fabrication detector	Flagged `gpt-5-mini` 5× more than `gpt-4o`; **67%** of flags are context-derivable (88.88 is 0.8888 as a %). Penalises showing your work.

Also: prompt sensitivity is **model-generation-specific**. Cycle-mean spread across four prompt strategies is 14.7/9.0 points for `gpt-4o` and **0.0**/**1.9** for `gpt-5-mini` on identical input, so "LLM auditors are unstable under paraphrase" needs a model and a date attached.

4 WHY IT MATTERS

The errors run in the dangerous direction. A false alarm costs reviewer time; a false clearance closes an open question. Both the floor result and the age result are false clearances — the direction least likely to prompt a second look, produced by an audit run competently on clean data.

A single benchmark number can measure task difficulty. Our headline model-agreement figure was 75%; it decomposes to 33%/92%/100%, and the 100% is the case where the audited classifier barely discriminates, so agreeing costs nothing. Any benchmark averaging over heterogeneous instances can report a number dominated by its difficulty mix.

Every mechanical proxy for judgement is a policy. A length threshold is a policy about what counts as an answer; a keyword list, about what counts as a remedy; a float match, about what counts as evidence. None was labelled as such, and each punished the better answer exactly where judgement mattered most. As evaluation automates, this failure class scales with it.

5 WHAT TO DO

State the favorable outcome explicitly and test it. **Print what the floor withheld**, with n and the unreported ratio. Bind protected-class definitions to the governing statute in code, with the citation beside the threshold. Audit the population, not a convenience sample. Record which reference group a ratio uses. Treat every gate as a hypothesis to falsify, starting with the terse answer and the tiny subgroup.

6 ARTIFACTS

Results `artifacts/consolidated/RESULTS.md`; paper `report/report_facct.tex`; floor analysis `scripts/compare_populations.py`. Every figure regenerates offline.